

NETWORK AUTOMATION



ANSIBLE



GNS3

OVERVIEW

What are GNS3 and Ansible

Workflow

Basic setup

Dynamic routing

WHAT IS ANSIBLE ?

- Ansible is an open-source automation tool used for IT orchestration, configuration management, application deployment, and task automation.
- Agentless – Uses SSH (Linux) or WinRM (Windows) to manage nodes—no need to install additional software on target systems.
- YAML-based – Uses simple, human-readable playbooks (.yml files) to define automation tasks.
- Idempotent – Ensures consistent results, even if a playbook runs multiple times.

ANSIBLE COMPONENTS

- Control Node: Hosts Ansible & runs commands, orchestrates tasks via SSH/WinRM.
- Managed Nodes: Remote systems controlled by Ansible, no agent required.
- Inventory: Lists managed nodes, groups systems logically.
- Modules: Reusable scripts for tasks , hundreds built-in; supports custom modules.

ANSIBLE COMPONENTS

- Playbooks: YAML files defining automation workflows, combine tasks, variables, and handlers.
- Plugins: Extend Ansible.
- Roles: Predefined, reusable playbook structures, simplify complex automation.

ANSIBLE INVENTORY

```
[routers]
R1 ansible_host=192.168.139.100
R2 ansible_host=192.168.139.200
R3 ansible_host=192.168.139.233

[routers:vars]
ansible_network_os=cisco.ios.ios
ansible_username=DUY
ansible_password=cisco
ansible_connection=network_cli
```

- INI (initialization) format: used to store configuration settings with simple structure
- Groups ([routers]) list managed nodes with IPs (ansible_host).
- Group variables ([group:vars]) define connection settings:
 - Network OS: set to match network platform
 - Credentials: username and password to log in to network devices
 - Connection: network_cli is a connection plugin in Ansible utilizes SSH to establish remote connections

ANSIBLE PLAYBOOK

```
— Home
- name: My first cisco playbook ( Show how
  hosts: routers
  gather_facts: false

  tasks:
    - name: Show uptime of cisco ios router
      ios_command:
        command: show version | i uptime
      register: output
    - name: display output
      debug:
        var: output.stdout_lines
```

Purpose: Check Cisco router uptime

Key Parts:

- Targets: routers group
- Task 1: Run show version command
- Task 2: Display uptime results

Format: YAML with indentation

WHAT IS GNS3 ?

- GNS3, short for Graphical Network Simulator-3, is a free and open-source network emulator used for designing, simulating, and testing network topologies.
- Network Simulation: GNS3 enables the creation of virtual networks, mimicking real-world network setups.
- Virtual Devices: It uses software images of network devices, such as Cisco IOS, to simulate their physical behavior.
- No Physical Hardware Needed: Build and test networks without the expense and complexity of physical routers and switches.

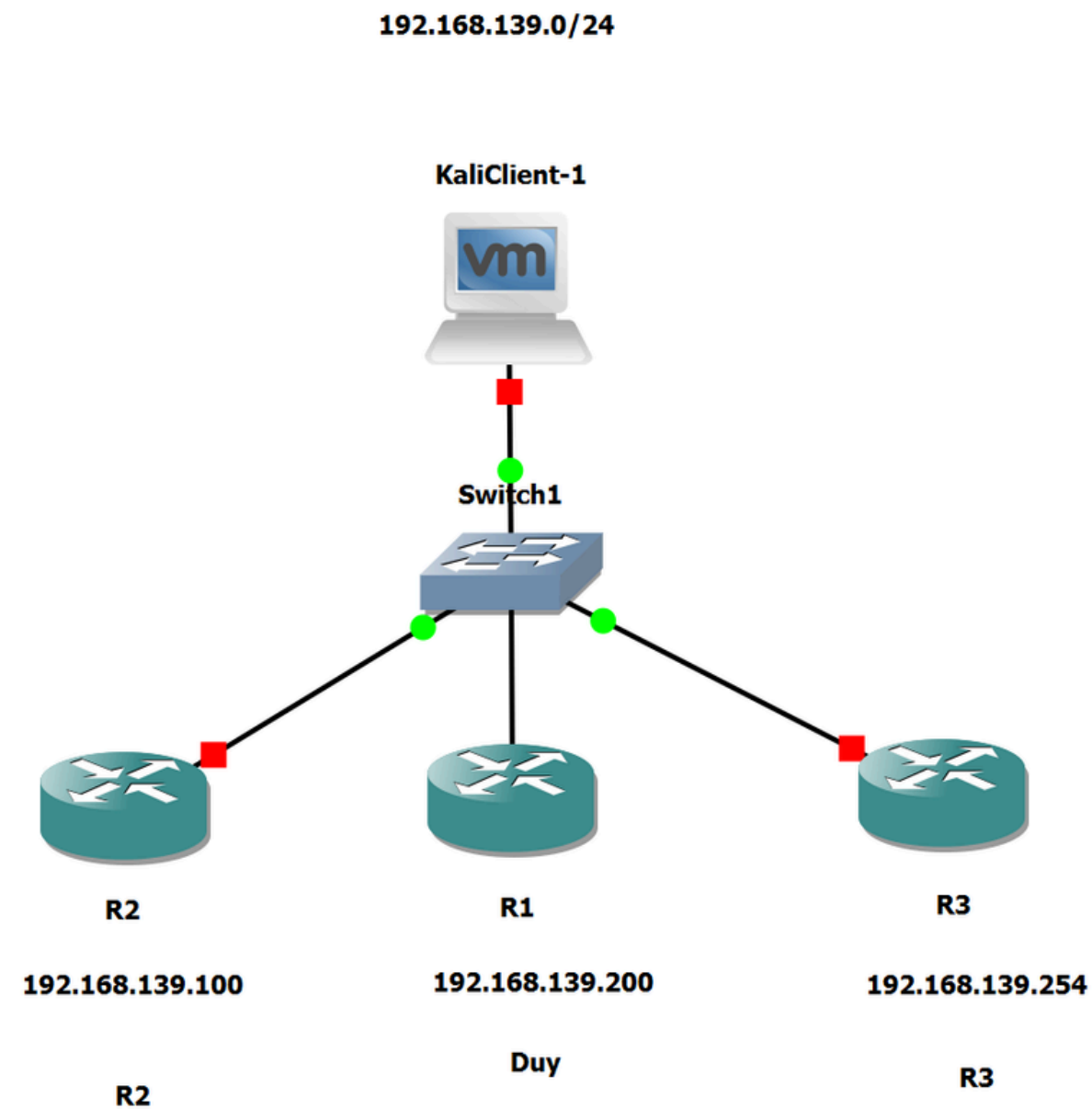
GNS3 ARCHITECTURE

- Host (The Control Center): Runs the GNS3 GUI.
- GNS3 VM (The Powerhouse): A virtual machine (Ubuntu) that runs the actual routers/switches., uses CPU's power to emulate devices. Without this, PC might freeze from overload!
- Emulated Devices (The Toys): Real OS Images: Cisco IOS, Juniper vSRX (run in QEMU/Dynamips).
Containers: Lightweight devices (Docker).
- Connections (The Cables): Connect devices like physical cables. Link lab to the internet or host.

WORKFLOW

- Setup: Install GNS3 All-in-One + GNS3 VM (VMware/VirtualBox).
- Design: Build topology in GNS3 GUI.
- Run: Launch devices and config basic settings
- Automate: Use Ansible to push configs via SSH

BASIC SETUP



Create a scenario with 3 cisco 3725 routers connect directly to a switch that connect to the control host (kali VM)

The network in this scenario is 192.168.139.0/24, configure all 3 routers with the IP address in range 192.168.139.1 - 192.168.139.254.

Now all devices have the same network as the control host.

BASIC SETUP

Create inventory for the devices to manage them.

```
#A collection of hosts belonging to the "routers" group:
[routers]
R1 ansible_host=192.168.139.200 ansible_network_os=ios ansible_username=DUY ansible_password=class ansible_connection=network_cli
R2 ansible_host=192.168.139.100 ansible_network_os=ios ansible_username=R2 ansible_password=class ansible_connection=network_cli
R3 ansible_host=192.168.139.254 ansible_network_os=ios ansible_username=R3 ansible_password=class ansible_connection=network_cli

#Previously, I used different names on each router so I cannot use group variables so I configured one by one
```

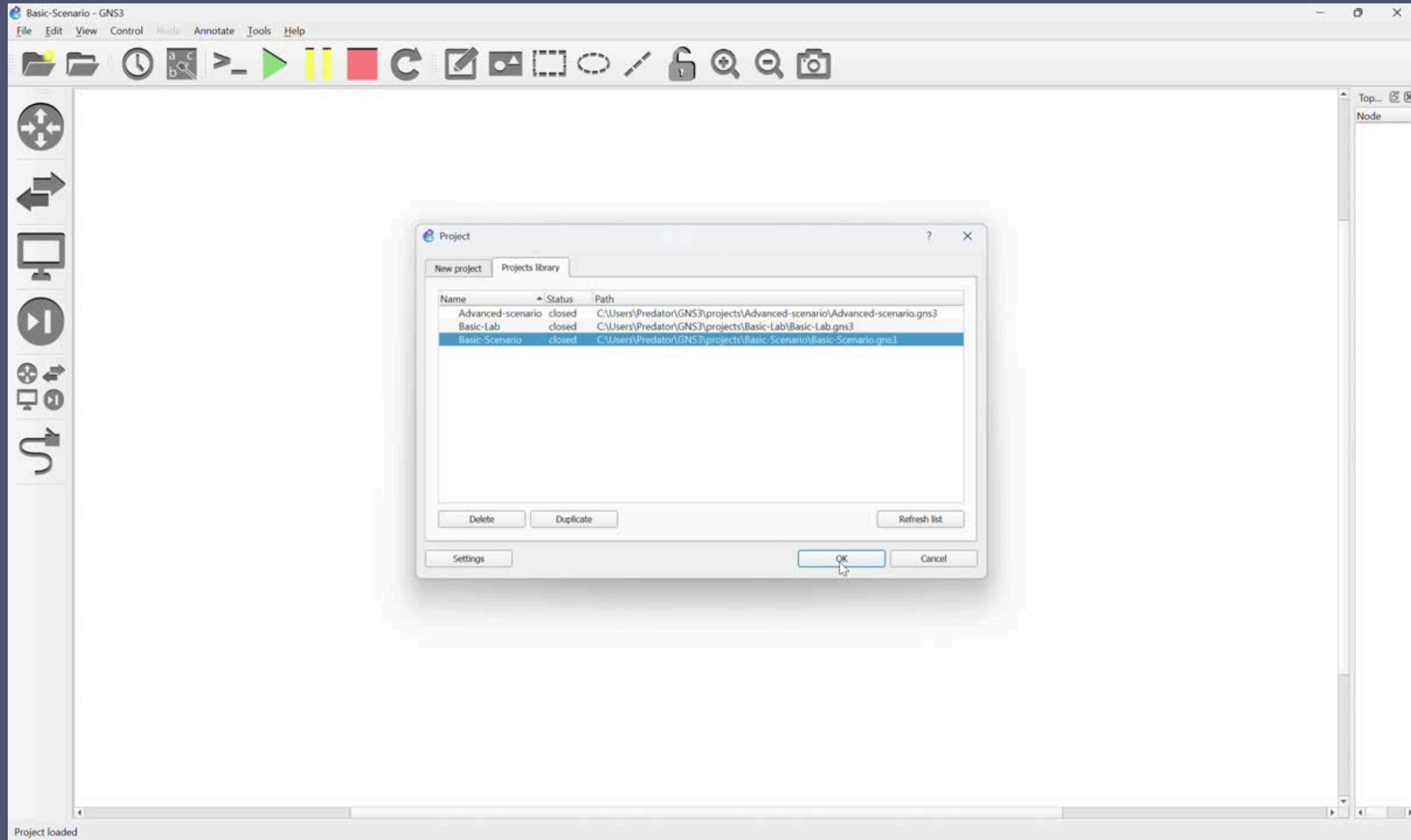
BASIC SETUP

Create playbook to do the automation

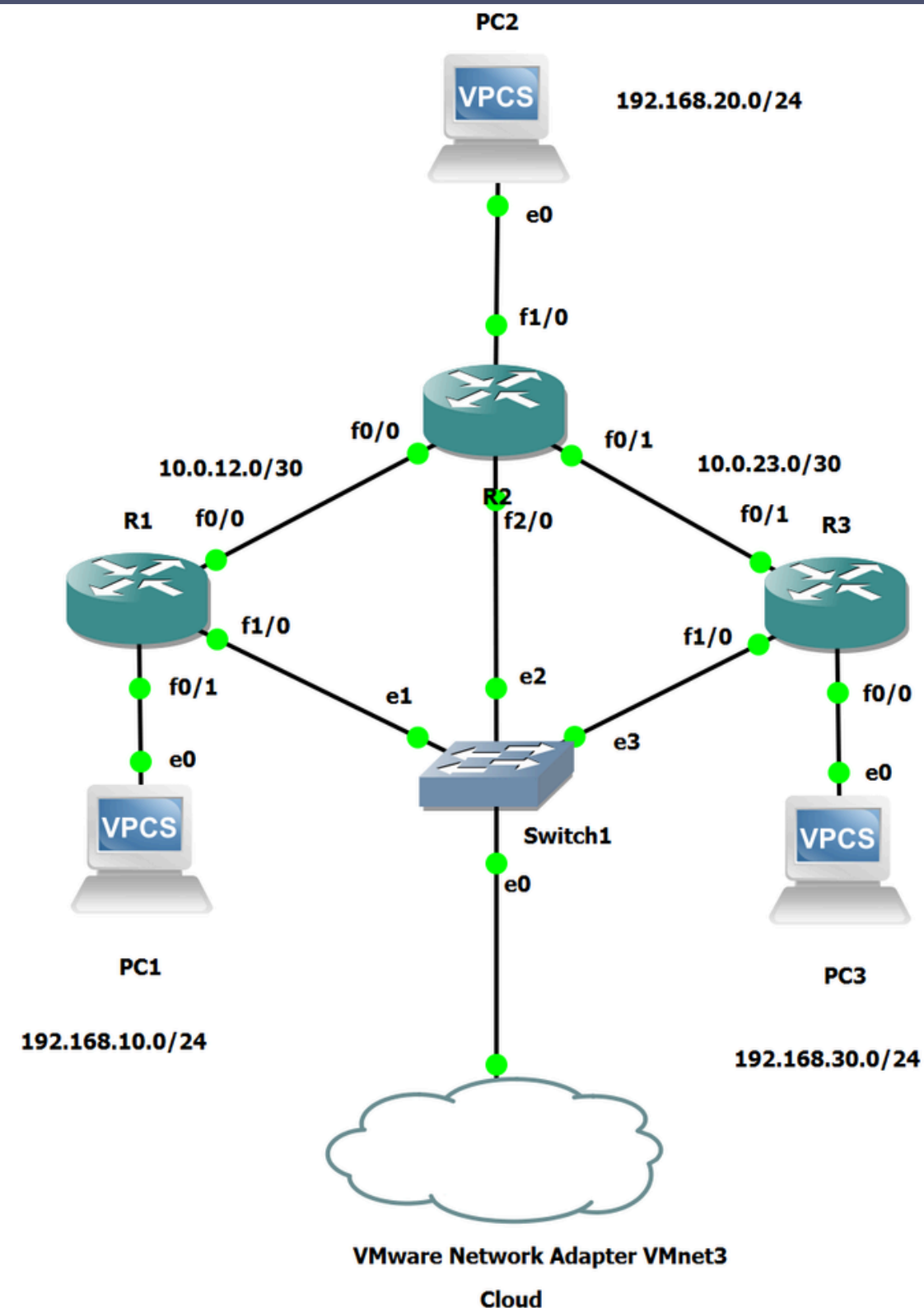
```
— Home
- name: My first cisco playbook ( Show how long the routers have been up )
  hosts: routers
  gather_facts: false

  tasks:
    - name: Show uptime of cisco ios router
      ios_command:
        command: show version | i uptime
      register: output
    - name: display output
      debug:
        var: output.stdout_lines
```

BASIC SETUP DEMO



DYNAMIC ROUTING



Create a scenario with 3 cisco 3725 routers connect directly to a switch that connect to the cloud which the cloud is assigned with virtual NIC in host machine

From each router, connect one VPCS (virtual PCs) to test connectivity. Then assign IP address for each router interface and VPCS.

Now the interface of the routers that connect to the cloud can route the traffic to outside internet.

DYNAMIC ROUTING

Create inventory for the devices to manage them.

```
[routers]
R1 ansible_host=192.168.139.100
R2 ansible_host=192.168.139.200
R3 ansible_host=192.168.139.233

[routers:vars]
ansible_network_os=cisco.ios.ios
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ansible_connection=network_cli
```

DYNAMIC ROUTING

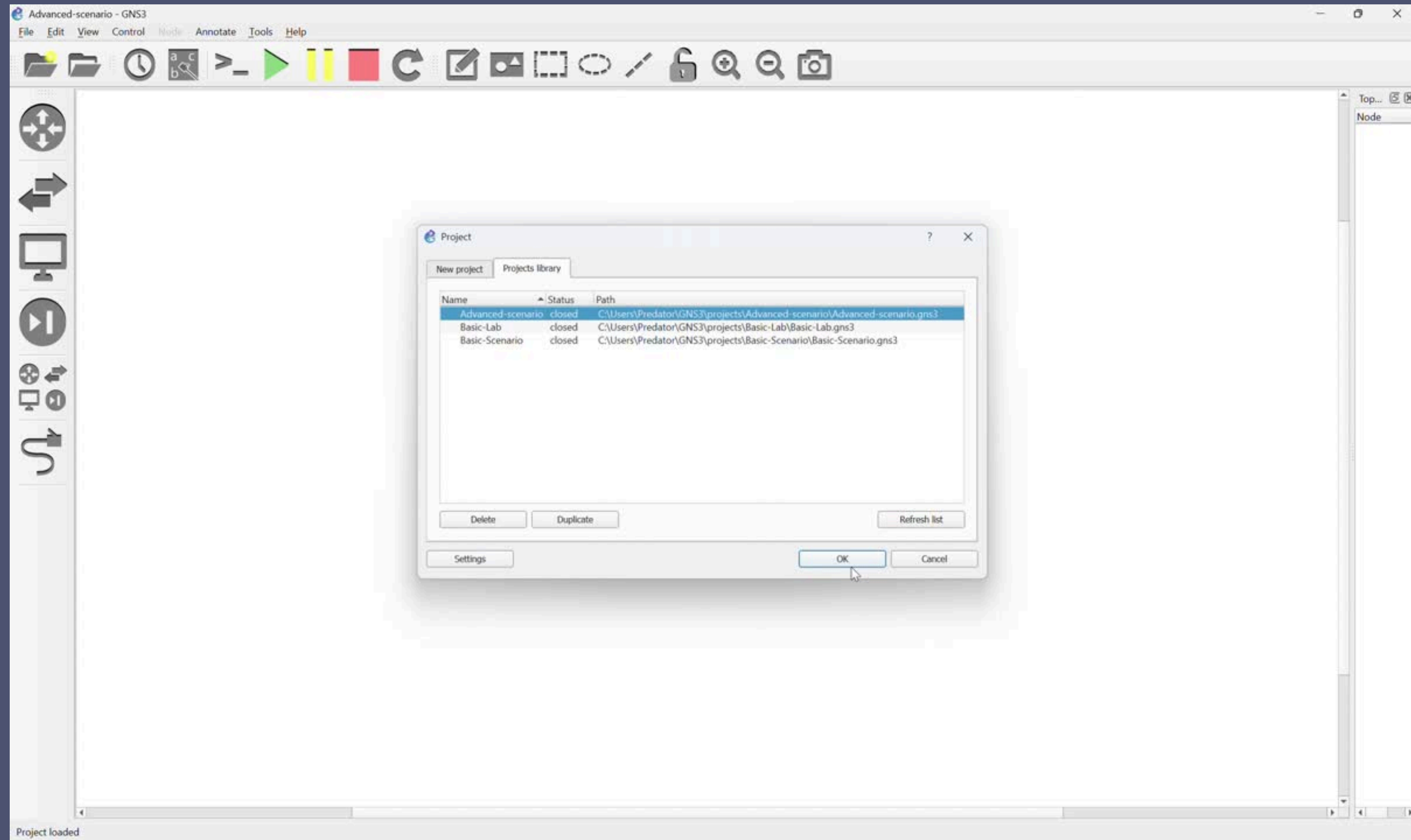
Create playbook to do dynamic routing auto configuration

```
Home
- name: Configure OSPF per router
  hosts: R1
  gather_facts: no
  tasks:
    - ios_config:
      lines:
        - router ospf 1
        - network 192.168.10.0 0.0.0.255 area 0
        - network 10.0.12.0 0.0.0.3 area 0

- name: Configure OSPF per router
  hosts: R2
  gather_facts: no
  tasks:
    - ios_config:
      lines:
        - router ospf 1
        - network 192.168.20.0 0.0.0.255 area 0
        - network 10.0.12.0 0.0.0.3 area 0
        - network 10.0.23.0 0.0.0.3 area 0
```

```
- name: Configure OSPF per router
  hosts: R3
  gather_facts: no
  tasks:
    - ios_config:
      lines:
        - router ospf 1
        - network 192.168.30.0 0.0.0.255 area 0
        - network 10.0.23.0 0.0.0.3 area 0
```

DYNAMIC ROUTING DEMO



Q&A

THANK YOU
FOR LISTENING!

